INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

REFRIGERATOR, MECHANICAL, BIOLOGICAL, LOW-TEMPERATURE

The General Services Administration has authorized the use of this Commercial Item Description in preference to MIL-R-3860.

1. ABSTRACT

1.1 Abstract. This commercial item description covers low-temperature, biological, mechanical refrigerators.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The biological refrigerator shall be a completely self-contained chest type, air cooled refrigerated unit with single top lid, capable of maintaining any selected storage compartment temperature between -5°F and +5°F, in an ambient of 90°F, ± 2°F. The refrigerated storage area shall be 12 cubic feet ± 1 cubic foot. Overall dimensions shall be 60-1/2 inches to 67 inches long, 28-1/2 inches to 32 inches wide, and 39 inches to 42 inches high. The exterior, interior, and lid of the cabinet shall be fabricated of 300 series stainless steel. The exterior of the cabinet including the lid shall have a minimum thickness of 18 (0.0480) gauge. The interior of the cabinet shall have a minimum thickness of 20 (0.0355) gauge. The interior of the lid shall have a minimum thickness of 16 (0.0595) gauge. The space between the exterior and interior

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AMSC N/A FSC 4110

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of the cabinet, and the lid shall be completely filled with insulation. Any blown-in insulation used during manufacturing shall use non-ozone depleting substances for propelling insulation. Loose type insulation shall not be used. Insulation shall be of sufficient amount so that no condensation will form on the exterior of the cabinet. The lid shall be furnished with a built-in locking device and hinges shall hold the lid in a raised position when open. The hinges shall be of the type that cannot be removed or opened when the lid is closed and locked. The cabinet shall be equipped with a dial-type thermometer which shall be flush mounted on the outside front of the refrigerator in a visible location. The cabinet shall be furnished with a minimum of four full swivel-type casters that have hard rubber or canvas-phenolic resin impregnated wheels. A minimum of two casters shall be equipped with a foot operated braking device which shall hold the cabinet stationary. Unless otherwise specified, the refrigerator shall be equipped with a battery operated light and buzzer set to give warning signals when the cabinet temperature rises above the temperature set on the temperature control (see 5.2). The warning devices shall be capable of being manually de-energized when the cabinet is not in use. A means shall be provided for checking the operation of the warning device without interrupting the operation of the refrigerator.

2.2 Refrigeration unit. The refrigeration unit shall be an integral part of the complete refrigerator. The assembled refrigeration unit shall be completely self-contained, completely wired, and completely dehydrated and charged with the proper amount and type of refrigerant and oil necessary for operation.

2.2.1 Temperature control. The refrigeration unit shall have a thermostatic-type temperature control. The temperature sensing element shall be located to be readily accessible and shall be easily removable. The control shall be adjustable to maintain temperatures selected between $-5^{\circ}F$ and $+5^{\circ}F$.

2.2.2 Refrigerant. The refrigeration system shall be charged with a commercially available, non-ozone depleting refrigerant. The oil used in the compressor shall be suitable for use with the refrigerant.

2.2.3 Ozone depleting substances. Class I ozone depleting substances, in accordance with section 602(a) of the Clean Air Act Amendments 1990, 42 USC 7671a(a) as implemented by 40 CFR 82, Protection of Stratospheric Ozone, shall not be used for refrigerant gases, or for any other purpose, or employed in any process associated with the manufacture of any component parts for this item. See the 5.5 for additional information.

2.3 Electrical requirements. Unless otherwise specified, the refrigeration unit shall operate on a nominal 115 V, 60 Hz, single-phase electrical system (see 5.2). A six foot minimum power cord made from a three-wire cord conforming to Underwriters Laboratories (UL) Standard No. 62 with a National Electrical Manufacturers Association (NEMA) Standard WD-1 three prong grounding plug shall be provided.

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2.4 Standard compliance. The biological refrigerator shall comply with American Society for Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE) Standard No. 13, UL Standard Nos. 62 and 471, and NEMA Standard No. WD-1, as applicable (see 3.2).

2.4 Data nameplate ("Government unique" requirement). The refrigerators shall be furnished with a data nameplate in accordance with the applicable requirements of MIL-STD-130 except the requirements for: (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data nameplate shall be made of minimum 20 gauge corrosion-resisting metal and attached to the refrigerator by rivets, screws, or welding in such a manner as to meet the applicable NSF International sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved, or applied by photosensitive means:

Contract Number
National Stock Number
Procurement Instrument Identification Number
CID Based Part Identification Number (PIN)
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (List Only if Different from Manufacturer)
Manufacturer's Model and Serial Number
Government Approved Manual Number

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the refrigerator.

2.5 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

2.6 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable. The term "recovered materials" means materials which have been collected or recovered from solid waste and

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reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description unless otherwise specified.

2.7 Workmanship. The finished refrigerators shall be complete, clean, and free of fabrication marks, scratches, dents, sharp edges, sharp corners, burrs, slivers, or visible evidence of rust. Rivets, when used, shall be tight and the rivet heads and peened ends shall be round and concentric and in full contact with the joined metal surfaces. There shall be no empty rivet holes. Welds shall be clean, continuous, sound, and free of flux, pits, buildups, undercuts, and shall be ground so as to blend with the adjacent surfaces. Spot welds shall be free from pits, burn throughs, and flashes.

3. QUALITY ASSURANCE

3.1 Contractor certification. The contractor shall certify that the product offered meets the salient characteristics and requirements of this commercial item description and except for those Government unique requirements specified, conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace or that had successfully been delivered to the Government on a previous contract or purchase order. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Compliance. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the refrigerator conforms to the requirements of ASHRAE Standard No. 13, UL Standard Nos. 62 and 471, and NEMA Standard No. WD-1, as applicable.

3.2.1 Underwriters Laboratories, Inc. (UL). Acceptable evidence of meeting the requirements of UL Standard No. 62 and No. 471 shall be the UL listing mark, or a certified test report from a recognized independent laboratory acceptable to the Government indicating compliance with the applicable requirements of UL Standard No. 62 and No. 471.

3.2.2 National Electrical Manufacturers Association (NEMA). The contractor shall submit to the contracting officer a written certificate of compliance as proof of compliance that the refrigerator meets the applicable requirements of the NEMA.

3.2.3 American Society for Heating, Refrigerating and Air Conditioning Engineers (ASHRAE). The contractor shall submit to the contracting officer a written certificate of compliance as proof that the biological refrigerators meet the applicable requirements of ASHRAE.

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4. PACKAGING

4.1 Packaging and packing. Packaging and packing shall be commercial or export as specified (see 5.2).

4.2 Commercial packaging and packing. Each refrigerator shall be preserved, packaged, and packed in accordance with ASTM D 3951.

4.3 Export packaging and packing.

4.3.1 Export packaging. Each refrigerator shall be preserved and cushioned to protect against corrosion, deterioration, and physical damage during shipment and storage.

4.3.2 Export packing. Each refrigerator shall be packed in a snug-fitting, wood-cleated plywood box conforming to PPP-B-601. The contents of each box shall be cushioned, blocked, and braced to provide a minimum of 1/2 inch clearance between the casters and the base of the box and to prevent movement and damage during transit. Each box shall be closed and strapped in accordance with PPP-B-601.

4.4 Marking. Marking of shipping containers shall be in accordance with MIL-STD-129.

5. NOTES

5.1 Intended use. The biological refrigerators covered by this commercial item description are intended to store blood serums, to preserve surgical and autopsy specimens, and to maintain in a frozen state, tissues from which virus isolation is attempted.

5.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this commercial item description.
b. When warning light and buzzer is not required (see 2.2).
c. Electrical requirements if other than specified (see 2.3).
d. Government approved manual number to be included on data nameplate (see 2.5).
e. Selection of applicable packaging and packing required (see 4.1).

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5.3 Sources for documents.

5.3.1 Sources for non-Government association documents.

ASHRAE Standard No. 13 - Methods of Rating and Testing Home Freezers

ASHRAE documents are available from:

American Society of Heating, Refrigerating, and
Air Conditioning Engineers, Inc.
United Engineering Center
345 East 47th Street
New York, NY 10017

ASTM D 3951 - Standard Practice for Commercial Packaging

ASTM documents are available from:

American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

NEMA Standard No. WD-1 - General requirements for Wiring Devices

NEMA documents are available from:

National Electrical Manufacturers Association (NEMA)
2101 L Street, NW
Washington, DC 20037

UL Standard No. 62 - UL Standard for Safety Flexible Cord and Fixture Wire
UL Standard No. 471 - UL Standard for Safety Commercial Refrigerators and Freezers

UL Standards are available from:

Underwriters Laboratories, Inc. (UL)
333 Pfingsten Road
Northbrook, IL 60062

5.3.2 Source for Government documents. Copies of military and federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

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5.4 "Government unique" requirements. Whenever a "Government unique" requirement is included in the title of a paragraph under "Salient Characteristics", it is meant that the requirement is something that is not normally offered to the commercial marketplace by the manufacturer.

5.5 Ozone depleting substances. For current information about the status of the Environmental Protection Agency's (EPA) Significant New Alternatives Policy (SNAP) program and specific refrigerant alternatives, call the Stratospheric Ozone Protection Hotline at (800) 296-1996 or the SNAP Coordinator at (202) 233-9195. For published information see the EPA Stratospheric Ozone Protection, Fact Sheet (EPA-430-F-93-004 dated February 1993) Short List of Alternative Refrigerants and Action Guide (EPA-430-F-93-003, dated February 1993) Cooling & Refrigerating Without CFCs.

MILITARY INTERESTS:

Custodians

Army - GL
Navy - SA
Air Force - 99

Review Activities

Army - MD
Air Force - 50
DLA - GS

User Activities

Army - MC, MS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

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